

73-103 · CMU-Q

Principles of Macroeconomics

Fall 2026

Andrés Castaño Zuluaga
Carnegie Mellon University Qatar
Business Administration

Labor Markets

Course roadmap

	Topic	Description	Toolkit	In the world
Measurement	GDP	What the economy produced	National accounts; $Y = C + I + G + NX$	Read a GDP release; split oil vs non-oil
	Prices	How to measure prices in a given economy	CPI and GDP deflator; real vs nominal	Index a wage; judge a currency peg
	Labor Markets	Employment, unemployment, participation	Labor-force identities; u and LFPR; labor demand and supply and their determinants ; institutions	Read a labor survey; GCC hiring rules
Short-run demand	Goods market	What determines demand? Equilibrium and the multiplier.	Goods-market equilibrium; multiplier	Size a fiscal package after an oil shock
	Financial markets	How i is set	Financial markets; Money market; the policy-rate rule;	Understand implications of a Fed rate decision
	IS-LM	Goods + financial together	IS and LM together; the policy mix	Map shocks into effects on output and the interest rate
Medium run	Phillips	Inflation and unemployment together	Short-run Phillips curve	Trade off inflation vs jobs in a boom
	Growth	What explain long term differences in economic growth and prosperity between countries differ	Production function; growth accounting	Explain why living standards differ
Open	Open economy	Trade, rates, and the peg	Net exports; peg vs float	Trace a shock through NX and the riyal peg

Midterm · Debate

Class roadmap

1 Discussion

Slides 4–8 · Mixed labor indicators, then AI and jobs by 2030

2 Measuring the labor force

Slides 9–11, 15 · Who is counted, the three rates, why official u is messy

3 Active Learning 1

Slides 12–14 · Compute LF, u , and participation from a table

4 Types of unemployment

Slides 16–19 · Natural rate, frictional search, structural mismatch

5 Wages and employment

Slides 20–23 · LD, LS, E at (L^*, W^*) , movement vs shifts

6 Four applications

Slides 24–27 · World Cup, GenAI, min wage, oil boom

7 Okun and Phillips

Slides 28–30 · Unemployment with output and with inflation

8 Active Learning 2

Slides 31–33 · Which policy reduces frictional unemployment?

9 Takeaways

Slides 34–35 · Types, equilibrium, and the two empirical links

SECTION 01

Discussion

Labor Market Indicators Point in All Directions

Peterson Institute for International Economics (PIIE) | January 16, 2026

The U.S. labor market currently presents a puzzle: some indicators are at their strongest levels in 25 years, while others suggest weakness rarely seen outside recessions.

Indicator	Current Status	What It Reveals
Prime-Age Emp-Pop Ratio	Historically strong	More working-age adults have jobs than in decades
Real Wages	Near all-time highs	Workers' purchasing power is increasing
Hiring Rate	Lowest since 2012	Employers are reluctant to add new workers
Monthly Payroll Growth	Weak	Job creation has slowed significantly

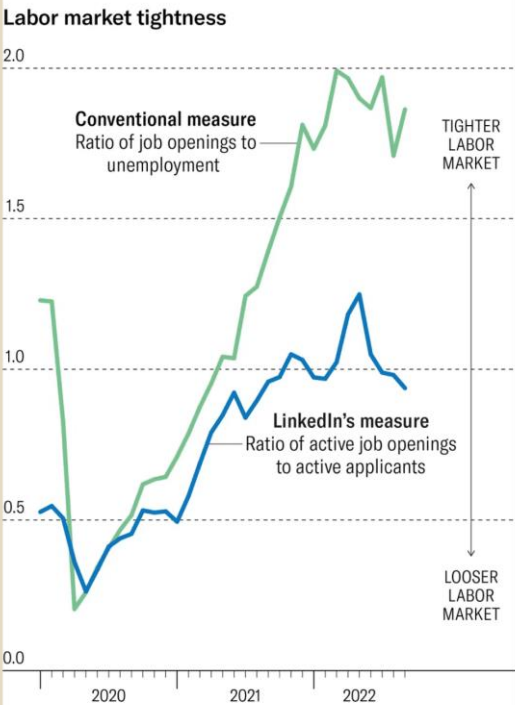
Key Insight

No single indicator tells the complete story. The unemployment rate alone can mask important dynamics in the labor market.

 [Read the article here](#)

How Tight Is the U.S. Labor Market?

LinkedIn's new measure of labor market slack suggests the labor market is not as tight as traditional measures indicate.



Source: LinkedIn, Job Openings and Labor Turnover Survey (JOLTS), Steven J. Davis, et al. 

Discussion Question 2

Analyzing Labor Market Indicators

According to the article, why is it important to look at multiple labor market indicators rather than relying solely on the unemployment rate?

Your Task

- 1 Identify at least two indicators from the article that tell different stories about the current labor market.
- 2 Explain what each indicator reveals about the state of employment in the U.S.

Think About It

"How might policymakers respond differently if they focused only on the unemployment rate versus looking at the full picture?"

How AI Could Reshape Jobs by 2030

World Economic Forum | January 7, 2026



Key Insight

The outcome depends on policy choices, corporate strategies, and how quickly workers can adapt to new technologies.

 [Read the article here](#)

Scenario	AI Adoption	Labor Demand	Wages & Skills
Supercharged Progress	Rapid, transformative	New jobs emerge quickly	Wages rise with productivity. Requires AI fluency & creativity.
Age of Displacement	Fast, disruptive	Jobs lost faster than created	Wages decline for displaced. Urgent reskilling needed.
Co-Pilot Economy	Incremental, collaborative	Human-AI teams	Stable, gradual growth. Requires adaptability & AI tools.
Stalled Progress	Conservative, limited	Minimal change	Stagnant wages. Traditional skills remain dominant.

Discussion Question 3

AI and the Future of Work

Of the four scenarios presented in the article, which one do you think is most likely to occur, and why?

Your Task:

In your answer, consider how your chosen scenario would affect:

- Labor demand
- Wages
- The types of skills workers will need

Think About It

You may draw on examples from the article or your own observations about AI, in the way you use it, or you have seen other people use it.

SECTION 02

Measuring Unemployment and key labor market indicators

How Is the Labor Force Measured?

Bureau of Labor Statistics (US) | Planning and Statistics Authority (Qatar)

Every adult (16 and older) falls into one of three categories based on their employment status.

Category	Definition	Examples
Employed	Paid work (employees, self-employed) or unpaid family business; temporarily absent (vacation, illness)	Full-time worker, part-time employee, business owner
Unemployed	Not employed, available for work, actively sought employment in past 4 weeks	Job seeker, recently laid off worker
Not in Labor Force	Not employed and not seeking work	Full-time students, homemakers, retirees

Key Insight

The labor force includes only the employed and unemployed. Those not seeking work are excluded from unemployment calculations.

Three Essential Labor Market Measures

01 Labor Force (LF)

$$\text{LF} = \text{Employed} + \text{Unemployed}$$

02 Unemployment Rate (UR)

$$\text{UR} = (\text{Unemployed} / \text{Labor Force}) \times 100$$

03 Labor Force Participation Rate (LFPR)

$$\text{LFPR} = (\text{Labor Force} / \text{Adult Population}) \times 100$$

Question

Among these three measures, which one is most telling of economic conditions?

ACTIVE LEARNING 01

Calculating Labor Market Indicators

Active Learning 1

Labor Force Calculations

Hypothetical Economy Data

Total Adult Population (16+)

260,000,000

Employed

150,000,000

Unemployed

10,000,000

Calculate the following:

- 1 Calculate the **Labor Force** .
- 2 Calculate the **Unemployment Rate** .
- 3 Calculate the **Labor Force Participation Rate** .
- 4 If 5 million unemployed workers become discouraged and stop looking for work, what happens to the unemployment rate?

Active Learning 1

Solution

Part 1: Initial Calculations

Measure	Formula	Calculation & Result
Labor Force	Employed + Unemployed	150M + 10M = 160,000,000
Unemployment Rate	(Unemployed / LF) × 100	(10M / 160M) × 100 = 6.25%
LFPR	(LF / Adult Pop) × 100	(160M / 260M) × 100 = 61.5%

Part 2: Discouraged Workers Scenario

New Unemployed	10M - 5M = 5M
New Labor Force	150M + 5M = 155M
New Unemployment Rate	(5M / 155M) × 100 = 3.2%

Key Takeaway

The unemployment rate fell from 6.25% to 3.2%, but this does not reflect an improvement. 5 million people simply stopped looking for work. This illustrates why we need multiple indicators to better understand trends in labor markets.

Challenges in Measuring Unemployment

Distinguishing between unemployed and out of the labor force is difficult. Several groups complicate measurement.

Group	Challenge	Impact on Statistics
Discouraged Workers	Stopped looking due to lack of opportunities	Unemployment rate understates joblessness
Marginally Attached	Want work but haven't searched recently	Not counted as unemployed
Part-Time for Economic Reasons	Want full-time but can only find part-time	Counted as employed despite underemployment
Underground Economy	Work "off the books" to avoid taxes	May report as unemployed

SECTION 03

Types of Unemployment

Understanding the Natural Rate of Unemployment

Unemployment can be divided into categories based on its causes and duration.

Type	Duration	Cause	Example
Frictional	Short-term, voluntary	Time to match workers with jobs	Recent graduate searching for first job
Structural	Long-term, involuntary	Mismatch between skills and job requirements	Coal miner in a region shifting to renewable energy
Cyclical	Varies with business cycle	Economic downturns reduce demand	Workers laid off during recession

Natural Rate Formula

Natural Rate = Frictional + Structural Unemployment

Key Insight

"Natural" does not mean desirable. It represents the unemployment rate when the economy is at full employment, e.g., cyclical unemployment is zero.

Frictional Unemployment: The Search Process

Unemployment that results from the time it takes to match workers with jobs. **It is often** short-term and voluntary.

Why It Exists

- **Imperfect Information:**
Workers don't know about every opening; firms don't know every candidate.
- **Sectoral Shifts:**
Changes in the composition of demand among industries or regions.
- **Career Transitions:**
Workers leaving jobs to find better matches.



Frictional unemployment ends when a successful match is made.

Is It Bad?

Not necessarily. Some frictional unemployment is inevitable and even beneficial. It indicates workers are taking time to find jobs that best suit their skills, leading to higher productivity and wages.

Structural Unemployment: Skills Mismatch

Occurs when there is a fundamental mismatch between the skills workers possess and the requirements of available jobs.

Primary Causes

Technological Change

Automation and AI replace routine cognitive and manual tasks, making certain skills obsolete.

Globalization

Production of goods moves to countries with lower labor costs, displacing domestic workers.

Sectoral Shifts

Permanent changes in consumer demand (e.g., the shift from coal to renewable energy).



Automation increases productivity but can displace workers with specific skill sets.

Key Difference: Unlike frictional unemployment, structural unemployment often lasts longer because workers need time to retrain or relocate.

SECTION 04

Labor Supply and Demand

What Determines Wages and Employment?

Labor Supply

Determined by Workers. The curve is upward sloping: as wages rise, the opportunity cost of leisure increases, so people work more.

Labor Demand

Determined by Firms. The curve is downward sloping: firms hire workers up to the point where the Value of the Marginal Product of Labor (VMPL) equals the wage.

Equilibrium

The wage adjusts to balance supply and demand. At the equilibrium wage, the quantity of labor supplied equals the quantity of labor demanded.

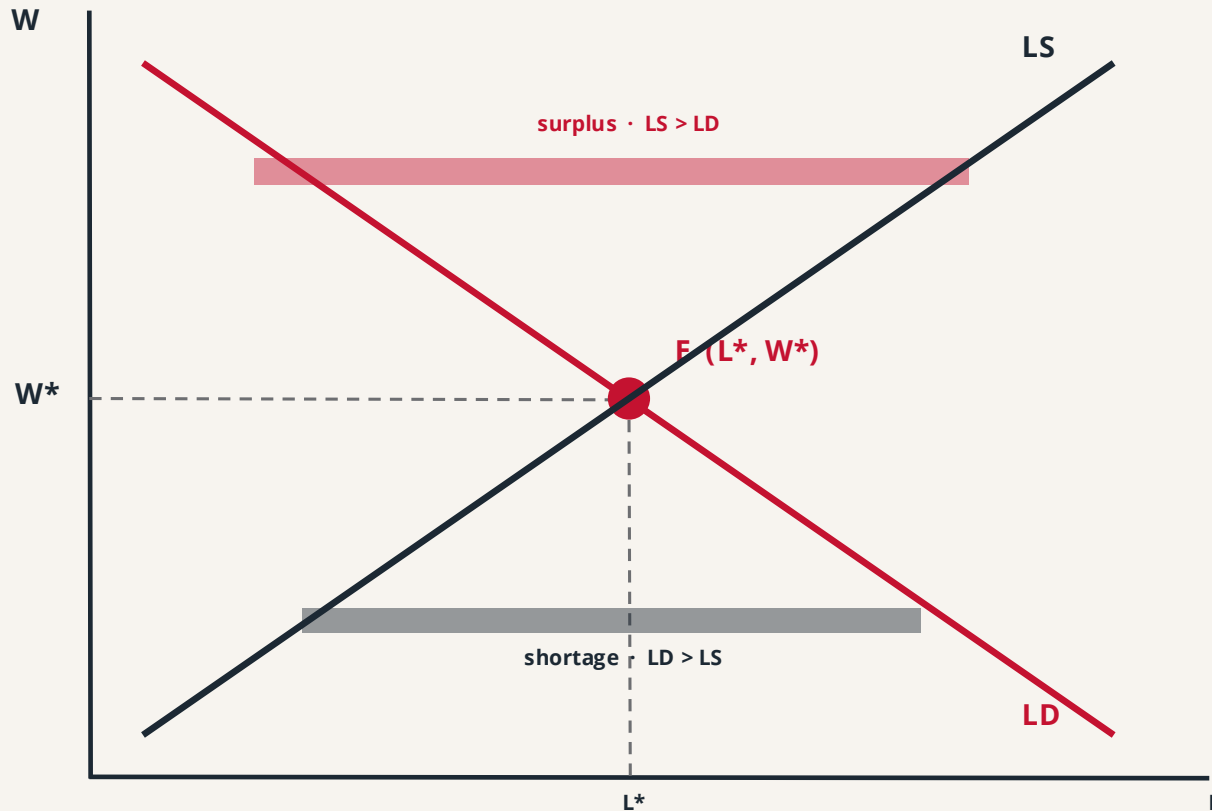


Key Insight

Any event that changes the supply or demand for labor must change the equilibrium wage and the value of the marginal product by the same amount.

Equilibrium in the Labor Market

Textbook figure · no scale · E at (L^*, W^*)



At $W > W^*$ there is a surplus of labor; at $W < W^*$ a shortage. The wage moves along both curves until $LD = LS$ at E .

MOVEMENT ALONG A CURVE

Only a change in the wage. Firms slide along LD ($VMPL = W$). Workers slide along LS (work vs leisure). L changes; the curves do not.

SHIFTS IN LABOR DEMAND

LD shifts when $VMPL$ changes — not when W changes.

- Output price / product demand
- Productivity or technology
- Prices of other inputs (complements vs substitutes)
- Number of firms

SHIFTS IN LABOR SUPPLY

LS shifts when non-wage factors change.

- Population and immigration
- Preferences and culture
- Alternative wages / school
- Amenities, safety, remote work

Determinants of Labor Demand and Labor Supply

Labor Demand (Firms)

- **Technology**
e.g., AI automates call centers, reducing demand for customer service agents.
- **Product Demand**
e.g., A surge in EV sales increases demand for battery engineers.
- **Prices of Other Inputs**
e.g., Cheaper industrial robots may substitute for assembly line workers.
- **Number of Firms**
e.g., More tech startups in a region increase aggregate demand for software developers.

Labor Supply (Workers)

- **Population Size**
e.g., Immigration policies in the Gulf increase the supply of construction workers.
- **Preferences & Culture**
e.g., Changing attitudes toward women in the workforce increase female labor supply.
- **Alternative Opportunities**
e.g., Higher university enrollment temporarily reduces the supply of young workers.
- **Non-Wage Factors**
e.g., Remote work options attract more applicants, increasing effective labor supply.

Qatar construction after the World Cup

Stadium and hotel work collapses in 2023–24. At the same moment Cabinet tightens new guest-worker visas for construction trades. Analyze the market for construction labor in Qatar.

1 Draw

Initial LD and LS. Mark $E_0 = (L_0, W_0)$.

Then draw the new curves after both events. Mark $E_1 = (L_1, W_1)$.

Label surplus or shortage if the wage is sticky for one year.

2 Which curve, and how?

For each event, say whether it is a movement along a curve or a shift of LD or LS — and the direction.

- A. End of stadium demand
- B. Tighter visas
- C. A later wage cut by contractors

3 Sign the changes

Give the sign of ΔL^* and ΔW^* — or write “ambiguous” and state the condition that would pin the wage down.

What would you need to know about the size of each shift?

Do not peek — then check the notes

Two shifters move at once. L^* is not ambiguous. W^* is. A wage cut by contractors is not a third shift.

Generative AI in call centers

A bank replaces 40% of routine call-center scripts with a chatbot. Remaining agents become “AI supervisors” whose VMPL rises. Bootcamps flood the market with new graduates who can do either job.

1 Draw two graphs

Market R: routine agents.
Market S: AI supervisors.

Start from a common W_0 . Show every shock that hits that market. Mark the new E in each.

2 Which function?

For each shock, name the market, the curve, and the direction.

- A. Chatbot substitutes for scripts
- B. Supervisors' VMPL rises
- C. Bootcamp graduates arrive

3 Relative wages

Must WR fall? Must WS rise? Can the skill premium WS / WR rise even if both wages fall?

State the condition in words.

Trap

The bootcamp is an LS shock in both markets if graduates can enter either job. Do not treat it as a demand shock.

A binding minimum wage, then a productivity boom

Retail starts at competitive E_0 . The state sets a minimum wage $W_m > W_0$. One year later, self-checkout and better logistics raise the VMPL of the workers who remain. The floor does not move.

1 Draw both stages

Stage I: LD, LS, W_m as a horizontal line. Mark $LD(W_m)$, $LS(W_m)$, and unemployment $U = LS - LD$.

Stage II: shift the correct curve. Remeasure U .

2 Which function?

- A. The statute that sets W_m
- B. The productivity boom
- C. Workers who quit because W_m is high

Which of these is a movement along LS?

3 Can U fall to zero?

Give the condition on the new VMPL schedule under which the floor stops binding.

If it still binds, does employment rise? Does measured unemployment have to fall?

Trap

The floor is not a shift of LD or LS. Employment can rise while U stays positive. U falls only if LD rises by more than LS does at W_m .

An oil-price boom in a Gulf labor market

Brent doubles. Energy firms bid aggressively for engineers and rig crews. Rents and grocery prices jump, so reservation wages rise in retail and hospitality. Guest-worker inflows are unchanged this year.

1 Draw two markets

Market E: energy labor.
Market N: non-oil services.

Show the oil shock in E. Show how higher living costs and poaching from E hit N. Mark both new equilibria.

2 Which function?

- A. Higher oil price (market E)
- B. Energy firms poach from N
- C. Higher rents raise reservation wages in N
- D. Unchanged visa quota

3 Who gains?

Sign ΔW and ΔL in E and in N.

Can non-oil employment fall while non-oil wages rise? That is the labor-market version of Dutch disease. Explain in one sentence.

Trap

Poaching is LS left in N (workers leave that market), not LD left. Higher reservation wages are also LS left. Do not shift LD in N unless product demand there changes.

SECTION 05

Output, Unemployment, and Inflation

Okun's Law and the Phillips Curve

Okun's Law: The Output-Unemployment Connection

There is a strong negative relationship between the unemployment rate and real GDP growth.

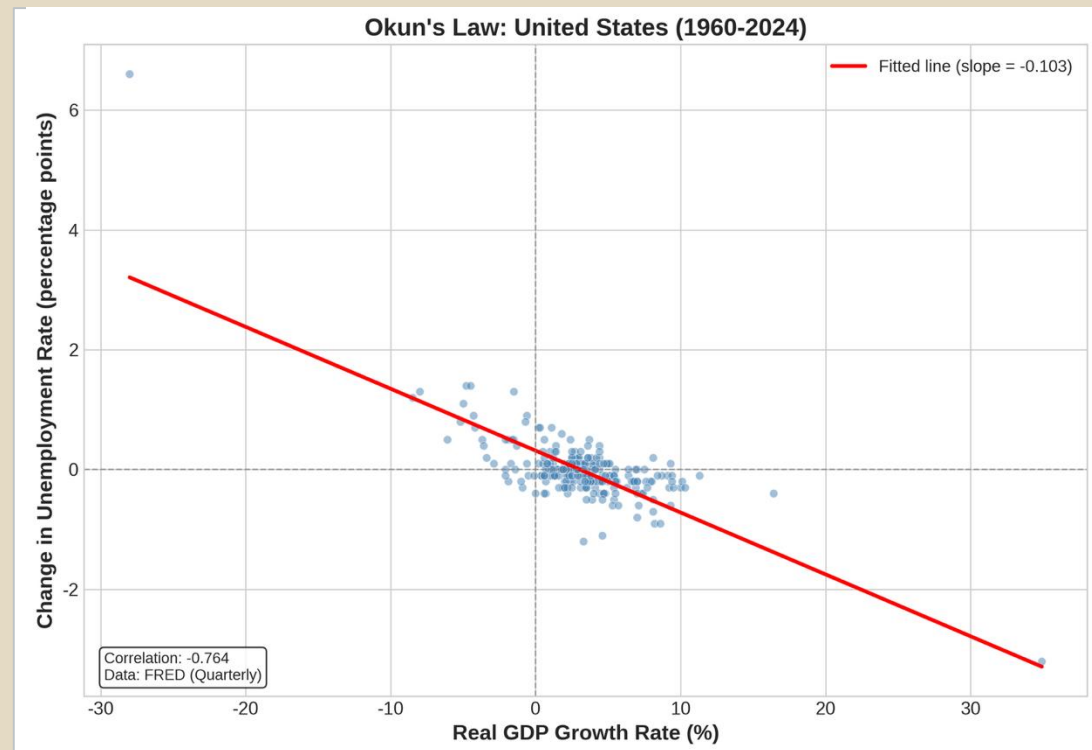
Rule of Thumb

For every **1% increase** in the unemployment rate, a country's real GDP is roughly **2% lower** than its potential GDP.

This means high unemployment is not just a social problem, but a waste of economic resources.

Key Insight

Because the labor force grows and productivity increases over time, the economy must grow (usually around 3%) just to keep the unemployment rate constant.



Historical data confirms that periods of high unemployment coincide with lower output growth.

The Phillips Curve: A Trade-Off?

The Phillips Curve shows the short-run inverse relationship between unemployment and inflation.

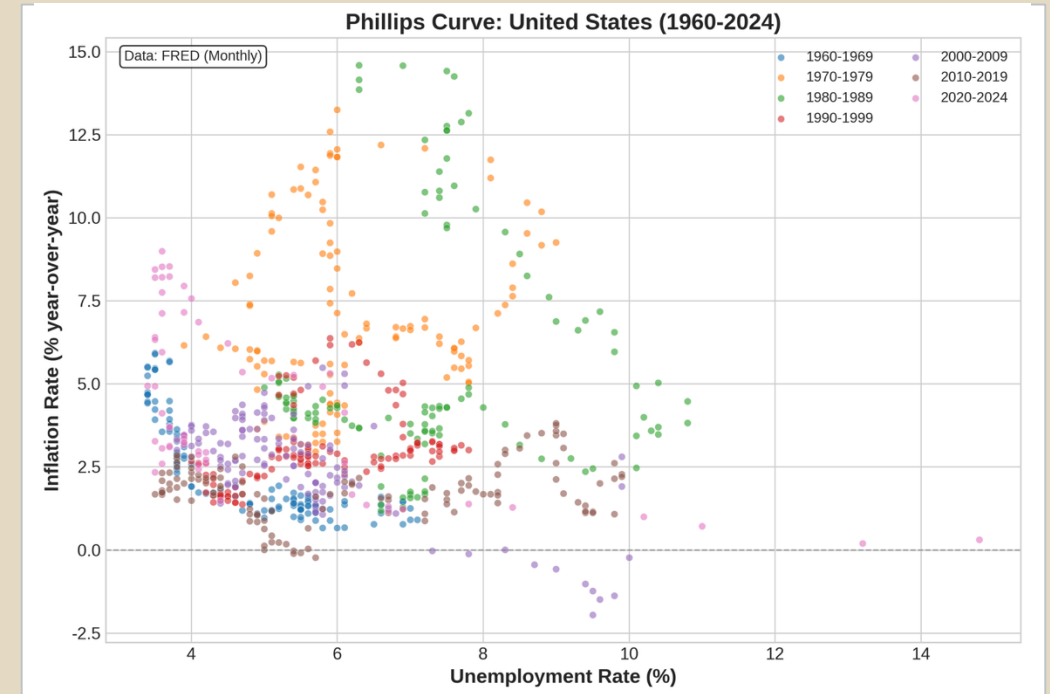
The Economic Logic

Low Unemployment → High Inflation

When labor demand is high, firms must offer higher wages to attract workers. Higher wages increase production costs, leading to higher prices.

High Unemployment → Low Inflation

When labor demand is low, wages stagnate or fall, reducing upward pressure on prices.



Crucial Distinction

This trade-off only holds in the short run. In the long run, the Phillips Curve is vertical at the natural rate of unemployment—inflation depends primarily on the money supply.

ACTIVE LEARNING 02

Reducing Frictional Unemployment

Policy Analysis

Active Learning 2

Policy Analysis

Which of the following government policies would reduce frictional unemployment?

Hint

Recall that frictional unemployment is caused by the time it takes for workers to search for jobs that match their skills.

A Increasing unemployment insurance benefits.

B Establishing government-run employment agencies to connect workers with jobs.

C Increasing the minimum wage.

D Passing laws that make it difficult to fire workers.

Active Learning 2

Solution

Correct Answer

B) Government-run employment agencies

Why is this correct?

Frictional unemployment is caused by the time it takes for workers to find jobs that match their skills and tastes.

Government employment agencies improve the flow of information about job openings, speeding up the matching process and reducing the duration of unemployment.

Why are the others incorrect?

Unemployment Insurance

Reduces the opportunity cost of being unemployed, which may actually increase the time workers spend searching (increasing frictional unemployment).

Minimum Wage Laws & Unions

These create a wage floor above the equilibrium level. This causes

structural unemployment

(quantity of labor supplied > quantity demanded), not frictional.

CONCLUSION

Key Takeaways

Key Takeaways

1 Three types

Frictional, structural, and cyclical unemployment have different causes and policies.

2 Measurement

Labor force, unemployment rate, and participation rate — and what each misses.

3 Wages and jobs

Labor demand and supply, determinant, and understanding equilibrium

4 Links and AI

Okun and Phillips connect output, unemployment, and inflation. AI is a labor-market shock, not a slogan.